

A Naive Bayes Classifier from Scratch for Heart Disease Prediction

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This project presents a Naive Bayes classifier implemented from scratch in Python and applied to the prediction of heart disease from clinical data. The classifier handles mixed data types within a single probabilistic framework, modeling categorical features with Laplace-smoothed discrete distributions and continuous features with class-conditional Gaussians. Computations are carried out in log-space for numerical stability, and missing values are handled natively by skipping their contribution to the likelihood rather than discarding or imputing records. We describe the underlying probabilistic model, the estimation of priors and class-conditional likelihoods, and the prediction rule, and evaluate the resulting classifier on a held-out test set.

I. INTRODUCTION

Naive Bayes classifiers are a family of probabilistic models built on a direct application of Bayes' theorem together with a strong conditional-independence assumption between features. Despite the simplicity of that assumption, they remain competitive in many practical settings, are inexpensive to train, and produce interpretable, probabilistic predictions, making them a useful baseline for classification problems.

In this project we implement a Naive Bayes classifier entirely from scratch, without relying on machine-learning libraries, and apply it to a heart disease dataset. The task is to predict a patient's disease category from a mix of clinical measurements, some categorical (such as chest-pain type and resting ECG results) and some continuous (such as age, cholesterol, and maximum heart rate). This mixture of feature types, along with the presence of missing values, motivates several of the modeling choices described below: discrete features are handled with Laplace smoothing, continuous features with Gaussian likelihoods, and missing entries are simply omitted from the per-instance likelihood. The dataset and its features are described in TABLE I, and the sections that follow develop the model and report its performance.

II. NAIVE BAYES CLASSIFIER FOR MIXED DATA

Given a dataset with class variable $Y \in \{1, \dots, k\}$ and feature vector $\mathbf{x} = (x_1, x_2, \dots, x_d)$, the posterior probability is

$$P(Y = y | \mathbf{x}) \propto P(\mathbf{x} | Y = y)P(Y = y) = \prod_{i=1}^d P(x_i | Y = y)P(Y = y)$$

The Naive Bayes classifier assigns a new instance to the class

$$\hat{y} = \arg \max_{y \in \mathcal{Y}} \left[P(Y = y) \prod_{i=1}^d P(x_i | Y = y) \right].$$

$$\hat{y} = \arg \max_{y \in \mathcal{Y}} \left[\log P(Y = y) + \sum_{i=1}^d \log P(x_i | Y = y) \right].$$

The fundamental assumption is that features are *conditionally independent* given the class.

Class Prior

The prior probability of class y is estimated from the training data as

$$P(Y = y) = \frac{n_y + \alpha}{n_{\text{tot}} + k\alpha},$$

where n_{tot} is the total number of training instances and n_y is the number of instances belonging to class y .

TABLE I. Description of Heart Disease Dataset Columns

Feature	Description
age	Age of the patient (in years).
sex	Gender of the patient ('male $\rightarrow$ 1', 'female $\rightarrow$ 0').
chest-pain	Type of chest pain experienced: 'abnang $\rightarrow$ 0' (atypical angina), 'angina $\rightarrow$ 1' (typical angina), 'asympt $\rightarrow$ 2' (asymptomatic), 'notang $\rightarrow$ 3' (non-anginal pain).
resting-blood-pressure	Resting blood pressure in mm Hg.
serum-cholesterol	Serum cholesterol in mg/dl.
fasting-blood-sugar	Fasting blood sugar > 120 mg/dl: '1 = true', '0 = false'.
rest-ecg	Resting electrocardiographic results: 'abn $\rightarrow$ 0' (ST-T wave abnormality), 'hyp $\rightarrow$ 1' (probable or definite left ventricular hypertrophy), 'norm $\rightarrow$ 2' (normal).
max-heart-rate	Maximum heart rate achieved during exercise.
exercise-angina	Exercise-induced angina: '1 = yes', '0 = no'.
oldpeak-ecg	ST depression induced by exercise relative to rest.
slope-ecg	Slope of the peak exercise ST segment: 'down $\rightarrow$ 0' (downsloping), 'flat $\rightarrow$ 1' (flat), 'up $\rightarrow$ 2' (upsloping).
major-vessels	Number of major vessels (0–3) colored by fluoroscopy.
thal	Thalassemia: 'fix $\rightarrow$ 0' (fixed defect), 'norm $\rightarrow$ 1' (normal), 'rev $\rightarrow$ 2' (reversible defect).
H, S1, S2, S3, S	One-hot encoding of the output class (disease category). Example: 'H=0, S1=0, S2=1, S3=0, S=0' indicates class 'S2'.

Discrete Attributes

Let attribute x_i be categorical with possible values $\{c_1, c_2, \dots, c_m\}$. For each class y , we estimate the conditional probabilities using *Laplace smoothing*. For each feature x_i we have

$$P(c_i | Y = y) = \frac{n_{iy} + \alpha}{n_y + m\alpha},$$

where

- n_{iy} = number of training instances in class y with c_i ,
- n_y = number of non-missing observations of x_i in class y ,
- α = smoothing parameter (typically $\alpha = 1$),
- m = number of distinct categories of attribute i .

If x_i is missing, the term $\log P(x_i | Y = y)$ is *skipped*.

Continuous Attributes

If attribute x_i is continuous, it is modeled using a Gaussian distribution:

$$P(x_i | Y = y) = \frac{1}{\sqrt{2\pi\sigma_{iy}^2}} \exp\left(-\frac{(x_i - \mu_{iy})^2}{2\sigma_{iy}^2}\right),$$

where

$$\mu_{iy} = \frac{1}{N_{iy}} \sum_{x_i \in D_{iy}} x_i, \quad \sigma_{iy}^2 = \frac{1}{N_{iy}} \sum_{x_i \in D_{iy}} (x_i - \mu_{iy})^2.$$

Here D_{iy} is the set of non-missing values of attribute i in class y . A small variance floor (e.g. $\sigma_{iy}^2 \leftarrow \max(\sigma_{iy}^2, 10^{-6})$) is used to avoid numerical issues.

If x_i is missing, the Gaussian term is skipped.

Prediction

For a new instance $\mathbf{x}$, we compute for each class y :

$$\log P(Y = y) + \sum_{i \in \text{observed}} \log P(x_i | Y = y),$$

and predict the class with the highest value.

III. RESULTS

Dataset	Accuracy
Training Set (80%)	66.5%
Testing Set (20%)	63.9%

TABLE II. Performance of the Naive Bayes Algorithm

IV. DISCUSSION

The classifier achieves an accuracy of 66.5% on the training set and 63.9% on the test set. The small gap between the two indicates that the model is not overfitting; if anything, both figures point to a degree of underfitting, which is consistent with the strong assumptions built into the method.

Several factors help explain the modest performance. The central limitation is the conditional-independence assumption itself: clinical features such as blood pressure, cholesterol, and maximum heart rate are in reality correlated, so treating them as independent given the class discards genuine structure in the data. The Gaussian model for continuous features is a further approximation, since not all of these variables are truly normally distributed within each class. The task is also inherently difficult, involving several disease categories rather than a simple binary healthy/diseased split, and the dataset is relatively small, which limits the quality of the estimated priors and class-conditional distributions, particularly for under-represented classes. Despite these limitations, the model behaves as expected for a Naive Bayes classifier and provides a transparent, fully interpretable baseline.